Lakshya JEE 2.0 (2025)

Physics

DPP: 6

Electric Charges and Fields

Q1 An electron having charge e and mass m is moving in a uniform electric field E. Its acceleration will be

(A) $\frac{e^2}{m}$
(B) $\frac{E^2 e}{m}$
(C) $\frac{eE}{m}$
(D) $\frac{mE}{e}$

Q2 Cathode rays travelling from east to west enter into region of electric field directed towards north to south in the plane of paper. The deflection of cathode rays is towards

(A) East (B) South
(C) West (D) North

Q3 A charged particle of mass m and charge q is released from rest in a uniform electric field E. Neglecting the effect of gravity, the kinetic energy of the charged particle after t second is

(A) $\frac{Eq^2 m}{2t^2}$
(B) $\frac{2E^2 t^2}{mq}$
(C) $\frac{E^2 q^2 t^2}{2m}$
(D) $\frac{Eqm}{t}$

Q4 The electric field on charge Q at the centroid of an equilateral triangle carrying an equal charge q at each of the vertices is

(A) Zero
(B) $\frac{\sqrt{2kq}}{r^2}$
(C) $\frac{kq}{\sqrt{2r^2}}$
(D) $\frac{3kq}{r^2}$

Q5 Charge $q_1 = +6.0nC$ is on Y-axis at $y = +3\ \mathrm{cm}$ and charge $q_2 = -6.0nC$ is on Y-axis at $y = -3\ \mathrm{cm}$ calculate force on a test charge $q_0 = 2nC$ placed on X-axis at $x = 4\ \mathrm{cm}$

(A) $-51.8\hat{j}\mu\mathrm{N}$
(B) $+51.8\hat{j}\mu\mathrm{N}$
(C) $-5.18\hat{j}\mu\mathrm{N}$
(D) $5.18\hat{j}\mu\mathrm{N}$

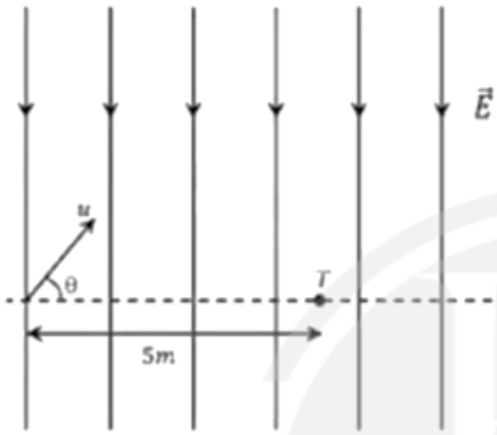
Q6 Under the influence of the coulomb field of charge $+Q$, a charge $-q$ is moving around it in an elliptical orbit. Find out the correct statements(s).

(A) The angular momentum of the charge $-q$ is constant
(B) The linear momentum of the charge $-q$ is constant
(C) The angular velocity of the charge $-q$ is constant
(D) The linear speed of the charge $-q$ is constant

Q7 An oil drop of radius $2\ \mathrm{mm}$ with a density $3\ \mathrm{g\ cm^{-3}}$ is held stationary under a constant electric field $3.55 \times 10^5 \mathrm{Vm^{-1}}$ in the Millikans oil drop experiment. What is the number of excess electrons that the oil drop will possess?
(consider $g = 9.81\ \mathrm{m/s^2}$)

(A) 48.8×10^{11}
(B) 1.73×10^{10}
(C) 17.3×10^{10}
(D) 1.73×10^{12}

Q8 A uniform electric field, $\overrightarrow{\mathrm{E}} = -400\sqrt{3}\hat{y}\mathrm{NC}^{-1}$ is applied in a region. A charged particle of mass m carrying positive charge q is projected in this egion with an initial speed of $2\sqrt{10} \times 10^6\ \mathrm{ms}^{-1}$. This particle is aimed to hit a target T, which is $5\ \mathrm{m}$ away from its entry point into the field as shown schematically in the figure. Take $\frac{\mathrm{q}}{\mathrm{m}} = 10^{10}\mathrm{Ckg}^{-1}$. Then



(A) The particle will hit if projected at an angle 45° from the horizontal

(B) The particle will hit T if projected either at an angle 30° or 60° from the horizontal

(C) Time taken by the particle to hit T could be $\sqrt{\frac{5}{6}}\,\mu\mathrm{s}$ as well as $\sqrt{\frac{5}{2}}\,\mu\mathrm{s}$

(D) Time taken by the particle to hit T is $\sqrt{\frac{5}{3}}\,\mu\mathrm{s}$

Q9 A particle of mass $20\ \mathrm{g}$ and charge $-3\ \mathrm{mC}$ moving with a speed $40\ \mathrm{m/s}$ enters a region of electric field $80\ \mathrm{N/C}$ in the same direction of the electric field. Speed of the particle after 3 seconds is:

(A) $76\ \mathrm{m/s}$ (B) $4\ \mathrm{m/s}$

(C) $40\ \mathrm{m/s}$ (D) $12\ \mathrm{m/s}$

Answer Key

Q1	(C)
Q2	(D)
Q3	(C)
Q4	(A)
Q5	(A)
Q6	(A)
Q7	(B)
Q8	(B, C)
Q9	(B)

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